

1 Instances (CPO)

Table 1: Results for CFS (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
1 20 3 10 rep 1.txt	20	3	10	Solution	30.09	755	631.00	16.42
2 20 3 10 rep 2.txt	20	3	10	Solution	30.02	740	678.00	8.38
3 20 3 10 rep 3.txt	20	3	10	Solution	30.04	498	465.00	6.63
4 20 3 10 rep 4.txt	20	3	10	Solution	30.08	742	668.00	9.97
5 20 3 10 rep 5.txt	20	3	10	Solution	30.01	880	864.00	1.82
6 20 3 10 rep 6.txt	20	3	10	Solution	30.04	726	618.00	14.88
7 20 3 10 rep 7.txt	20	3	10	Solution	30.01	719	668.00	7.09
8 20 3 10 rep 8.txt	20	3	10	Solution	30.01	881	766.00	13.05
9 20 3 10 rep 9.txt	20	3	10	Solution	30.02	599	563.00	6.01
10 20 3 10 rep 10.txt	20	3	10	Solution	30.03	784	716.00	8.67
11 20 3 20 rep 1.txt	20	3	20	Optimal	5.15	412	412.00	0.00
12 20 3 20 rep 2.txt	20	3	20	Optimal	2.24	358	358.00	0.00
13 20 3 20 rep 3.txt	20	3	20	Solution	30.01	421	409.00	2.85
14 20 3 20 rep 4.txt	20	3	20	Optimal	13.12	385	385.00	0.00
15 20 3 20 rep 5.txt	20	3	20	Optimal	6.13	350	350.00	0.00
16 20 3 20 rep 6.txt	20	3	20	Optimal	2.08	383	383.00	0.00
17 20 3 20 rep 7.txt	20	3	20	Solution	30.01	372	367.00	1.34
18 20 3 20 rep 8.txt	20	3	20	Optimal	12.83	413	413.00	0.00
19 20 3 20 rep 9.txt	20	3	20	Optimal	9.35	297	297.00	0.00
20 20 3 20 rep 10.txt	20	3	20	Optimal	2.01	353	353.00	0.00
21 20 3 30 rep 1.txt	20	3	30	Optimal	2.55	335	335.00	0.00
22 20 3 30 rep 2.txt	20	3	30	Optimal	6.21	296	296.00	0.00
23 20 3 30 rep 3.txt	20	3	30	Optimal	5.34	304	304.00	0.00
24 20 3 30 rep 4.txt	20	3	30	Optimal	0.49	292	292.00	0.00
25 20 3 30 rep 5.txt	20	3	30	Optimal	2.00	292	292.00	0.00
26 20 3 30 rep 6.txt	20	3	30	Optimal	2.13	293	293.00	0.00
27 20 3 30 rep 7.txt	20	3	30	Optimal	6.74	345	345.00	0.00
28 20 3 30 rep 8.txt	20	3	30	Optimal	2.65	324	324.00	0.00
29 20 3 30 rep 9.txt	20	3	30	Optimal	4.14	300	300.00	0.00
30 20 3 30 rep 10.txt	20	3	30	Optimal	0.20	245	245.00	0.00
31 20 3 50 rep 1.txt	20	3	50	Optimal	0.02	253	253.00	0.00
32 20 3 50 rep 2.txt	20	3	50	Optimal	0.18	237	237.00	0.00
33 20 3 50 rep 3.txt	20	3	50	Optimal	0.02	241	241.00	0.00
34 20 3 50 rep 4.txt	20	3	50	Optimal	0.19	232	232.00	0.00
35 20 3 50 rep 5.txt	20	3	50	Optimal	0.02	272	272.00	0.00
36 20 3 50 rep 6.txt	20	3	50	Optimal	0.02	216	216.00	0.00
37 20 3 50 rep 7.txt	20	3	50	Optimal	0.02	224	224.00	0.00
38 20 3 50 rep 8.txt	20	3	50	Optimal	0.02	264	264.00	0.00
39 20 3 50 rep 9.txt	20	3	50	Optimal	0.05	234	234.00	0.00
40 20 3 50 rep 10.txt	20	3	50	Optimal	0.02	231	231.00	0.00
41 20 5 10 rep 1.txt	20	5	10	Solution	30.02	946	829.00	12.37
42 20 5 10 rep 2.txt	20	5	10	Solution	30.02	916	777.00	15.17

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Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
43 20 5 10 rep 3.txt	20	5	10	Solution	30.03	929	800.00	13.89
44 20 5 10 rep 4.txt	20	5	10	Solution	30.03	917	737.00	19.63
45 20 5 10 rep 5.txt	20	5	10	Solution	30.02	915	767.00	16.17
46 20 5 10 rep 6.txt	20	5	10	Solution	30.04	832	683.00	17.91
47 20 5 10 rep 7.txt	20	5	10	Solution	30.01	1011	759.00	24.93
48 20 5 10 rep 8.txt	20	5	10	Solution	30.01	957	754.00	21.21
49 20 5 10 rep 9.txt	20	5	10	Solution	30.01	900	783.00	13.00
50 20 5 10 rep 10.txt	20	5	10	Solution	30.01	1100	946.00	14.00
51 20 5 20 rep 1.txt	20	5	20	Solution	30.03	497	489.00	1.61
52 20 5 20 rep 2.txt	20	5	20	Optimal	12.91	441	441.00	0.00
53 20 5 20 rep 3.txt	20	5	20	Solution	30.01	518	505.00	2.51
54 20 5 20 rep 4.txt	20	5	20	Solution	30.01	387	381.00	1.55
55 20 5 20 rep 5.txt	20	5	20	Optimal	14.93	478	478.00	0.00
56 20 5 20 rep 6.txt	20	5	20	Solution	30.04	508	491.00	3.35
57 20 5 20 rep 7.txt	20	5	20	Solution	30.02	479	435.00	9.19
58 20 5 20 rep 8.txt	20	5	20	Optimal	11.34	511	511.00	0.00
59 20 5 20 rep 9.txt	20	5	20	Optimal	3.85	522	522.00	0.00
60 20 5 20 rep 10.txt	20	5	20	Solution	30.01	486	428.00	11.93
61 20 5 30 rep 1.txt	20	5	30	Optimal	3.06	387	387.00	0.00
62 20 5 30 rep 2.txt	20	5	30	Optimal	5.14	452	452.00	0.00
63 20 5 30 rep 3.txt	20	5	30	Optimal	0.14	367	367.00	0.00
64 20 5 30 rep 4.txt	20	5	30	Solution	30.03	430	421.00	2.09
65 20 5 30 rep 5.txt	20	5	30	Optimal	1.36	385	385.00	0.00
66 20 5 30 rep 6.txt	20	5	30	Optimal	0.90	402	402.00	0.00
67 20 5 30 rep 7.txt	20	5	30	Optimal	0.02	431	431.00	0.00
68 20 5 30 rep 8.txt	20	5	30	Optimal	1.00	404	404.00	0.00
69 20 5 30 rep 9.txt	20	5	30	Optimal	10.47	441	441.00	0.00
70 20 5 30 rep 10.txt	20	5	30	Optimal	17.48	437	437.00	0.00
71 20 5 50 rep 1.txt	20	5	50	Optimal	0.02	312	312.00	0.00
72 20 5 50 rep 2.txt	20	5	50	Optimal	0.06	323	323.00	0.00
73 20 5 50 rep 3.txt	20	5	50	Optimal	0.02	353	353.00	0.00
74 20 5 50 rep 4.txt	20	5	50	Optimal	0.02	348	348.00	0.00
75 20 5 50 rep 5.txt	20	5	50	Optimal	0.02	371	371.00	0.00
76 20 5 50 rep 6.txt	20	5	50	Optimal	0.02	425	425.00	0.00
77 20 5 50 rep 7.txt	20	5	50	Optimal	0.02	349	349.00	0.00
78 20 5 50 rep 8.txt	20	5	50	Optimal	0.02	398	398.00	0.00
79 20 5 50 rep 9.txt	20	5	50	Optimal	0.02	398	398.00	0.00
80 20 5 50 rep 10.txt	20	5	50	Optimal	0.02	386	386.00	0.00
81 20 10 10 rep 1.txt	20	10	10	Solution	30.02	1261	964.00	23.55
82 20 10 10 rep 2.txt	20	10	10	Solution	30.01	1272	1000.00	21.38
83 20 10 10 rep 3.txt	20	10	10	Solution	30.01	1303	1002.00	23.10
84 20 10 10 rep 4.txt	20	10	10	Solution	30.02	1290	1119.00	13.26
85 20 10 10 rep 5.txt	20	10	10	Solution	30.01	1362	1056.00	22.47
86 20 10 10 rep 6.txt	20	10	10	Solution	30.02	1275	1070.00	16.08
87 20 10 10 rep 7.txt	20	10	10	Solution	30.04	1238	1018.00	17.77

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Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
88 20 10 10 rep 8.txt	20	10	10	Solution	30.03	1302	1082.00	16.90
89 20 10 10 rep 9.txt	20	10	10	Solution	30.02	1238	987.00	20.27
90 20 10 10 rep 10.txt	20	10	10	Solution	30.02	1251	984.00	21.34
91 20 10 20 rep 1.txt	20	10	20	Solution	30.01	759	735.00	3.16
92 20 10 20 rep 2.txt	20	10	20	Solution	30.04	852	735.00	13.73
93 20 10 20 rep 3.txt	20	10	20	Optimal	6.68	704	704.00	0.00
94 20 10 20 rep 4.txt	20	10	20	Solution	30.02	853	740.00	13.25
95 20 10 20 rep 5.txt	20	10	20	Solution	30.01	732	704.00	3.83
96 20 10 20 rep 6.txt	20	10	20	Solution	30.01	877	831.00	5.25
97 20 10 20 rep 7.txt	20	10	20	Solution	30.02	801	723.00	9.74
98 20 10 20 rep 8.txt	20	10	20	Solution	30.04	808	731.00	9.53
99 20 10 20 rep 9.txt	20	10	20	Solution	30.01	806	740.00	8.19
100 20 10 20 rep 10.txt	20	10	20	Optimal	22.43	747	747.00	0.00
101 20 10 30 rep 1.txt	20	10	30	Optimal	0.14	730	730.00	0.00
102 20 10 30 rep 2.txt	20	10	30	Optimal	0.10	780	780.00	0.00
103 20 10 30 rep 3.txt	20	10	30	Optimal	0.91	615	615.00	0.00
104 20 10 30 rep 4.txt	20	10	30	Optimal	0.24	726	726.00	0.00
105 20 10 30 rep 5.txt	20	10	30	Solution	30.03	668	662.00	0.90
106 20 10 30 rep 6.txt	20	10	30	Optimal	0.32	639	639.00	0.00
107 20 10 30 rep 7.txt	20	10	30	Optimal	0.20	728	728.00	0.00
108 20 10 30 rep 8.txt	20	10	30	Optimal	0.05	763	763.00	0.00
109 20 10 30 rep 9.txt	20	10	30	Optimal	0.80	645	645.00	0.00
110 20 10 30 rep 10.txt	20	10	30	Optimal	0.05	692	692.00	0.00
111 20 10 50 rep 1.txt	20	10	50	Optimal	0.02	668	668.00	0.00
112 20 10 50 rep 2.txt	20	10	50	Optimal	0.02	740	740.00	0.00
113 20 10 50 rep 3.txt	20	10	50	Optimal	0.02	701	701.00	0.00
114 20 10 50 rep 4.txt	20	10	50	Optimal	0.02	693	693.00	0.00
115 20 10 50 rep 5.txt	20	10	50	Optimal	0.02	651	651.00	0.00
116 20 10 50 rep 6.txt	20	10	50	Optimal	0.02	641	641.00	0.00
117 20 10 50 rep 7.txt	20	10	50	Optimal	0.02	654	654.00	0.00
118 20 10 50 rep 8.txt	20	10	50	Optimal	0.02	673	673.00	0.00
119 20 10 50 rep 9.txt	20	10	50	Optimal	0.02	708	708.00	0.00
120 20 10 50 rep 10.txt	20	10	50	Optimal	0.02	731	731.00	0.00
121 30 3 10 rep 1.txt	30	3	10	Solution	30.00	1005	886.00	11.84
122 30 3 10 rep 2.txt	30	3	10	Solution	30.04	1249	1120.00	10.33
123 30 3 10 rep 3.txt	30	3	10	Solution	30.01	1035	867.00	16.23
124 30 3 10 rep 4.txt	30	3	10	Solution	30.01	1202	1031.00	14.23
125 30 3 10 rep 5.txt	30	3	10	Solution	30.01	1105	996.00	9.86
126 30 3 10 rep 6.txt	30	3	10	Solution	30.01	1198	1069.00	10.77
127 30 3 10 rep 7.txt	30	3	10	Solution	30.03	1221	1123.00	8.03
128 30 3 10 rep 8.txt	30	3	10	Solution	30.01	1181	999.00	15.41
129 30 3 10 rep 9.txt	30	3	10	Solution	30.03	1055	972.00	7.87
130 30 3 10 rep 10.txt	30	3	10	Solution	30.02	1039	950.00	8.57
131 30 3 20 rep 1.txt	30	3	20	Solution	30.03	568	524.00	7.75
132 30 3 20 rep 2.txt	30	3	20	Solution	30.03	584	539.00	7.71

Table 1: Results for CFS (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
133 30 3 20 rep 3.txt	30	3	20	Solution	30.02	503	440.00	12.52
134 30 3 20 rep 4.txt	30	3	20	Solution	30.01	626	592.00	5.43
135 30 3 20 rep 5.txt	30	3	20	Solution	30.03	474	418.00	11.81
136 30 3 20 rep 6.txt	30	3	20	Solution	30.02	513	487.00	5.07
137 30 3 20 rep 7.txt	30	3	20	Solution	30.02	526	474.00	9.89
138 30 3 20 rep 8.txt	30	3	20	Solution	30.01	566	534.00	5.65
139 30 3 20 rep 9.txt	30	3	20	Solution	30.02	504	485.00	3.77
140 30 3 20 rep 10.txt	30	3	20	Solution	30.03	468	453.00	3.21
141 30 3 30 rep 1.txt	30	3	30	Solution	30.05	439	423.00	3.64
142 30 3 30 rep 2.txt	30	3	30	Solution	30.01	333	317.00	4.80
143 30 3 30 rep 3.txt	30	3	30	Solution	30.01	450	440.00	2.22
144 30 3 30 rep 4.txt	30	3	30	Solution	30.01	344	333.00	3.20
145 30 3 30 rep 5.txt	30	3	30	Solution	30.00	436	423.00	2.98
146 30 3 30 rep 6.txt	30	3	30	Solution	30.01	336	304.00	9.52
147 30 3 30 rep 7.txt	30	3	30	Solution	30.03	417	407.00	2.40
148 30 3 30 rep 8.txt	30	3	30	Solution	30.02	389	378.00	2.83
149 30 3 30 rep 9.txt	30	3	30	Solution	30.01	419	412.00	1.67
150 30 3 30 rep 10.txt	30	3	30	Solution	30.03	368	360.00	2.17
151 30 3 50 rep 1.txt	30	3	50	Optimal	0.11	252	252.00	0.00
152 30 3 50 rep 2.txt	30	3	50	Optimal	0.11	258	258.00	0.00
153 30 3 50 rep 3.txt	30	3	50	Optimal	0.03	294	294.00	0.00
154 30 3 50 rep 4.txt	30	3	50	Optimal	0.88	235	235.00	0.00
155 30 3 50 rep 5.txt	30	3	50	Optimal	23.11	253	253.00	0.00
156 30 3 50 rep 6.txt	30	3	50	Optimal	0.93	268	268.00	0.00
157 30 3 50 rep 7.txt	30	3	50	Optimal	0.85	253	253.00	0.00
158 30 3 50 rep 8.txt	30	3	50	Optimal	7.83	260	260.00	0.00
159 30 3 50 rep 9.txt	30	3	50	Optimal	0.03	275	275.00	0.00
160 30 3 50 rep 10.txt	30	3	50	Solution	30.00	267	258.00	3.37
161 30 5 10 rep 1.txt	30	5	10	Solution	30.04	1310	1053.00	19.62
162 30 5 10 rep 2.txt	30	5	10	Solution	30.01	1287	1093.00	15.07
163 30 5 10 rep 3.txt	30	5	10	Solution	30.02	1302	1091.00	16.21
164 30 5 10 rep 4.txt	30	5	10	Solution	30.01	1183	1008.00	14.79
165 30 5 10 rep 5.txt	30	5	10	Solution	30.01	1371	1173.00	14.44
166 30 5 10 rep 6.txt	30	5	10	Solution	30.01	1204	907.00	24.67
167 30 5 10 rep 7.txt	30	5	10	Solution	30.02	1267	1099.00	13.26
168 30 5 10 rep 8.txt	30	5	10	Solution	30.01	1680	1370.00	18.45
169 30 5 10 rep 9.txt	30	5	10	Solution	30.02	1071	878.00	18.02
170 30 5 10 rep 10.txt	30	5	10	Solution	30.01	1422	1179.00	17.09
171 30 5 20 rep 1.txt	30	5	20	Solution	30.00	686	581.00	15.31
172 30 5 20 rep 2.txt	30	5	20	Solution	30.04	638	585.00	8.31
173 30 5 20 rep 3.txt	30	5	20	Solution	30.01	699	605.00	13.45
174 30 5 20 rep 4.txt	30	5	20	Solution	30.01	703	631.00	10.24
175 30 5 20 rep 5.txt	30	5	20	Solution	30.03	588	533.00	9.35
176 30 5 20 rep 6.txt	30	5	20	Solution	30.03	679	645.00	5.01
177 30 5 20 rep 7.txt	30	5	20	Solution	30.00	673	600.00	10.85

Table 1: Results for CFS (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
178 30 5 20 rep 8.txt	30	5	20	Solution	30.01	643	544.00	15.40
179 30 5 20 rep 9.txt	30	5	20	Solution	30.04	623	535.00	14.13
180 30 5 20 rep 10.txt	30	5	20	Solution	30.02	696	655.00	5.89
181 30 5 30 rep 1.txt	30	5	30	Solution	30.00	462	429.00	7.14
182 30 5 30 rep 2.txt	30	5	30	Solution	30.02	480	444.00	7.50
183 30 5 30 rep 3.txt	30	5	30	Solution	30.01	490	445.00	9.18
184 30 5 30 rep 4.txt	30	5	30	Solution	30.04	483	453.00	6.21
185 30 5 30 rep 5.txt	30	5	30	Solution	30.03	515	463.00	10.10
186 30 5 30 rep 6.txt	30	5	30	Solution	30.00	414	393.00	5.07
187 30 5 30 rep 7.txt	30	5	30	Solution	30.01	476	447.00	6.09
188 30 5 30 rep 8.txt	30	5	30	Solution	30.01	472	461.00	2.33
189 30 5 30 rep 9.txt	30	5	30	Solution	30.02	479	449.00	6.26
190 30 5 30 rep 10.txt	30	5	30	Solution	30.04	517	472.00	8.70
191 30 5 50 rep 1.txt	30	5	50	Solution	30.03	368	366.00	0.54
192 30 5 50 rep 2.txt	30	5	50	Optimal	0.16	378	378.00	0.00
193 30 5 50 rep 3.txt	30	5	50	Optimal	5.45	379	379.00	0.00
194 30 5 50 rep 4.txt	30	5	50	Optimal	0.10	382	382.00	0.00
195 30 5 50 rep 5.txt	30	5	50	Optimal	3.64	367	367.00	0.00
196 30 5 50 rep 6.txt	30	5	50	Optimal	0.02	379	379.00	0.00
197 30 5 50 rep 7.txt	30	5	50	Optimal	1.12	394	394.00	0.00
198 30 5 50 rep 8.txt	30	5	50	Optimal	0.02	445	445.00	0.00
199 30 5 50 rep 9.txt	30	5	50	Optimal	1.21	360	360.00	0.00
200 30 5 50 rep 10.txt	30	5	50	Optimal	19.54	378	378.00	0.00
201 30 10 10 rep 1.txt	30	10	10	Solution	30.03	1828	1478.00	19.15
202 30 10 10 rep 2.txt	30	10	10	Solution	30.01	1542	1134.00	26.46
203 30 10 10 rep 3.txt	30	10	10	Solution	30.01	1655	1240.00	25.08
204 30 10 10 rep 4.txt	30	10	10	Solution	30.03	1784	1414.00	20.74
205 30 10 10 rep 5.txt	30	10	10	Solution	30.04	1641	1279.00	22.06
206 30 10 10 rep 6.txt	30	10	10	Solution	30.03	1778	1417.00	20.30
207 30 10 10 rep 7.txt	30	10	10	Solution	30.02	1666	1323.00	20.59
208 30 10 10 rep 8.txt	30	10	10	Solution	30.03	1650	1230.00	25.45
209 30 10 10 rep 9.txt	30	10	10	Solution	30.02	1737	1281.00	26.25
210 30 10 10 rep 10.txt	30	10	10	Solution	30.02	1836	1364.00	25.71
211 30 10 20 rep 1.txt	30	10	20	Solution	30.05	938	824.00	12.15
212 30 10 20 rep 2.txt	30	10	20	Solution	30.03	886	728.00	17.83
213 30 10 20 rep 3.txt	30	10	20	Solution	30.02	996	852.00	14.46
214 30 10 20 rep 4.txt	30	10	20	Solution	30.02	913	791.00	13.36
215 30 10 20 rep 5.txt	30	10	20	Solution	30.04	947	855.00	9.71
216 30 10 20 rep 6.txt	30	10	20	Solution	30.01	1017	887.00	12.78
217 30 10 20 rep 7.txt	30	10	20	Solution	30.01	1007	860.00	14.60
218 30 10 20 rep 8.txt	30	10	20	Solution	30.01	938	866.00	7.68
219 30 10 20 rep 9.txt	30	10	20	Solution	30.01	904	787.00	12.94
220 30 10 20 rep 10.txt	30	10	20	Solution	30.01	993	802.00	19.23
221 30 10 30 rep 1.txt	30	10	30	Solution	30.01	758	720.00	5.01
222 30 10 30 rep 2.txt	30	10	30	Solution	30.05	782	742.00	5.12

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Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
223 30 10 30 rep 3.txt	30	10	30	Solution	30.04	701	698.00	0.43
224 30 10 30 rep 4.txt	30	10	30	Solution	30.06	727	710.00	2.34
225 30 10 30 rep 5.txt	30	10	30	Solution	30.03	775	698.00	9.94
226 30 10 30 rep 6.txt	30	10	30	Solution	30.01	791	737.00	6.83
227 30 10 30 rep 7.txt	30	10	30	Solution	30.03	733	690.00	5.87
228 30 10 30 rep 8.txt	30	10	30	Optimal	0.63	791	791.00	0.00
229 30 10 30 rep 9.txt	30	10	30	Solution	30.01	762	708.00	7.09
230 30 10 30 rep 10.txt	30	10	30	Solution	30.04	800	754.00	5.75
231 30 10 50 rep 1.txt	30	10	50	Optimal	0.02	738	738.00	0.00
232 30 10 50 rep 2.txt	30	10	50	Optimal	0.08	646	646.00	0.00
233 30 10 50 rep 3.txt	30	10	50	Optimal	0.03	753	753.00	0.00
234 30 10 50 rep 4.txt	30	10	50	Optimal	0.08	652	652.00	0.00
235 30 10 50 rep 5.txt	30	10	50	Optimal	0.09	667	667.00	0.00
236 30 10 50 rep 6.txt	30	10	50	Optimal	0.03	687	687.00	0.00
237 30 10 50 rep 7.txt	30	10	50	Optimal	0.03	827	827.00	0.00
238 30 10 50 rep 8.txt	30	10	50	Optimal	0.02	692	692.00	0.00
239 30 10 50 rep 9.txt	30	10	50	Optimal	0.25	671	671.00	0.00
240 30 10 50 rep 10.txt	30	10	50	Optimal	0.03	764	764.00	0.00
241 50 3 10 rep 1.txt	50	3	10	Solution	30.02	1861	1623.00	12.79
242 50 3 10 rep 2.txt	50	3	10	Solution	30.02	1682	1473.00	12.43
243 50 3 10 rep 3.txt	50	3	10	Solution	30.01	1898	1586.00	16.44
244 50 3 10 rep 4.txt	50	3	10	Solution	30.03	1803	1686.00	6.49
245 50 3 10 rep 5.txt	50	3	10	Solution	30.01	1683	1504.00	10.64
246 50 3 10 rep 6.txt	50	3	10	Solution	30.02	1785	1539.00	13.78
247 50 3 10 rep 7.txt	50	3	10	Solution	30.01	1682	1548.00	7.97
248 50 3 10 rep 8.txt	50	3	10	Solution	30.03	1653	1436.00	13.13
249 50 3 10 rep 9.txt	50	3	10	Solution	30.03	1793	1665.00	7.14
250 50 3 10 rep 10.txt	50	3	10	Solution	30.01	1507	1340.00	11.08
251 50 3 20 rep 1.txt	50	3	20	Solution	30.01	789	717.00	9.13
252 50 3 20 rep 2.txt	50	3	20	Solution	30.04	889	845.00	4.95
253 50 3 20 rep 3.txt	50	3	20	Solution	30.02	828	764.00	7.73
254 50 3 20 rep 4.txt	50	3	20	Solution	30.01	785	712.00	9.30
255 50 3 20 rep 5.txt	50	3	20	Solution	30.01	825	735.00	10.91
256 50 3 20 rep 6.txt	50	3	20	Solution	30.03	867	797.00	8.07
257 50 3 20 rep 7.txt	50	3	20	Solution	30.02	898	814.00	9.35
258 50 3 20 rep 8.txt	50	3	20	Solution	30.02	834	781.00	6.35
259 50 3 20 rep 9.txt	50	3	20	Solution	30.03	819	760.00	7.20
260 50 3 20 rep 10.txt	50	3	20	Solution	30.02	794	735.00	7.43
261 50 3 30 rep 1.txt	50	3	30	Solution	30.01	580	539.00	7.07
262 50 3 30 rep 2.txt	50	3	30	Solution	30.03	668	637.00	4.64
263 50 3 30 rep 3.txt	50	3	30	Solution	30.01	538	502.00	6.69
264 50 3 30 rep 4.txt	50	3	30	Solution	30.05	539	510.00	5.38
265 50 3 30 rep 5.txt	50	3	30	Solution	30.04	584	563.00	3.60
266 50 3 30 rep 6.txt	50	3	30	Solution	30.01	564	529.00	6.21
267 50 3 30 rep 7.txt	50	3	30	Solution	30.02	614	603.00	1.79

Table 1: Results for CFS (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
268 50 3 30 rep 8.txt	50	3	30	Solution	30.04	556	535.00	3.78
269 50 3 30 rep 9.txt	50	3	30	Solution	30.03	617	590.00	4.38
270 50 3 30 rep 10.txt	50	3	30	Solution	30.01	598	566.00	5.35
271 50 3 50 rep 1.txt	50	3	50	Solution	30.04	364	346.00	4.95
272 50 3 50 rep 2.txt	50	3	50	Solution	30.01	365	355.00	2.74
273 50 3 50 rep 3.txt	50	3	50	Solution	30.02	352	333.00	5.40
274 50 3 50 rep 4.txt	50	3	50	Solution	30.01	380	372.00	2.11
275 50 3 50 rep 5.txt	50	3	50	Solution	30.03	355	335.00	5.63
276 50 3 50 rep 6.txt	50	3	50	Solution	30.02	381	364.00	4.46
277 50 3 50 rep 7.txt	50	3	50	Solution	30.03	377	368.00	2.39
278 50 3 50 rep 8.txt	50	3	50	Solution	30.02	333	326.00	2.10
279 50 3 50 rep 9.txt	50	3	50	Solution	30.03	380	359.00	5.53
280 50 3 50 rep 10.txt	50	3	50	Solution	30.03	357	350.00	1.96
281 50 5 10 rep 1.txt	50	5	10	Solution	30.03	1834	1657.00	9.65
282 50 5 10 rep 2.txt	50	5	10	Solution	30.02	2127	1846.00	13.21
283 50 5 10 rep 3.txt	50	5	10	Solution	30.01	1868	1525.00	18.36
284 50 5 10 rep 4.txt	50	5	10	Solution	30.03	2053	1742.00	15.15
285 50 5 10 rep 5.txt	50	5	10	Solution	30.02	2256	1770.00	21.54
286 50 5 10 rep 6.txt	50	5	10	Solution	30.02	2035	1823.00	10.42
287 50 5 10 rep 7.txt	50	5	10	Solution	30.02	1921	1724.00	10.26
288 50 5 10 rep 8.txt	50	5	10	Solution	30.03	1905	1545.00	18.90
289 50 5 10 rep 9.txt	50	5	10	Solution	30.02	2059	1773.00	13.89
290 50 5 10 rep 10.txt	50	5	10	Solution	30.03	2029	1697.00	16.36
291 50 5 20 rep 1.txt	50	5	20	Solution	30.02	995	908.00	8.74
292 50 5 20 rep 2.txt	50	5	20	Solution	30.01	978	849.00	13.19
293 50 5 20 rep 3.txt	50	5	20	Solution	30.02	948	855.00	9.81
294 50 5 20 rep 4.txt	50	5	20	Solution	30.01	1119	970.00	13.32
295 50 5 20 rep 5.txt	50	5	20	Solution	30.02	1063	962.00	9.50
296 50 5 20 rep 6.txt	50	5	20	Solution	30.01	974	884.00	9.24
297 50 5 20 rep 7.txt	50	5	20	Solution	30.02	931	881.00	5.37
298 50 5 20 rep 8.txt	50	5	20	Solution	30.03	963	856.00	11.11
299 50 5 20 rep 9.txt	50	5	20	Solution	30.01	948	854.00	9.92
300 50 5 20 rep 10.txt	50	5	20	Solution	30.03	910	869.00	4.51
301 50 5 30 rep 1.txt	50	5	30	Solution	30.04	672	597.00	11.16
302 50 5 30 rep 2.txt	50	5	30	Solution	30.01	660	599.00	9.24
303 50 5 30 rep 3.txt	50	5	30	Solution	30.04	670	585.00	12.69
304 50 5 30 rep 4.txt	50	5	30	Solution	30.01	694	611.00	11.96
305 50 5 30 rep 5.txt	50	5	30	Solution	30.03	683	628.00	8.05
306 50 5 30 rep 6.txt	50	5	30	Solution	30.02	696	653.00	6.18
307 50 5 30 rep 7.txt	50	5	30	Solution	30.01	717	621.00	13.39
308 50 5 30 rep 8.txt	50	5	30	Solution	30.04	744	720.00	3.23
309 50 5 30 rep 9.txt	50	5	30	Solution	30.02	686	609.00	11.22
310 50 5 30 rep 10.txt	50	5	30	Solution	30.01	724	678.00	6.35
311 50 5 50 rep 1.txt	50	5	50	Solution	30.04	440	401.00	8.86
312 50 5 50 rep 2.txt	50	5	50	Solution	30.04	486	459.00	5.56

Table 1: Results for CFS (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
313 50 5 50 rep 3.txt	50	5	50	Solution	30.01	468	445.00	4.91
314 50 5 50 rep 4.txt	50	5	50	Solution	30.01	455	419.00	7.91
315 50 5 50 rep 5.txt	50	5	50	Solution	30.04	471	439.00	6.79
316 50 5 50 rep 6.txt	50	5	50	Solution	30.02	421	397.00	5.70
317 50 5 50 rep 7.txt	50	5	50	Optimal	0.86	440	440.00	0.00
318 50 5 50 rep 8.txt	50	5	50	Solution	30.03	445	410.00	7.87
319 50 5 50 rep 9.txt	50	5	50	Solution	30.04	430	404.00	6.05
320 50 5 50 rep 10.txt	50	5	50	Solution	30.04	425	397.00	6.59
321 50 10 10 rep 1.txt	50	10	10	Solution	30.03	2446	1911.00	21.87
322 50 10 10 rep 2.txt	50	10	10	Solution	30.02	2377	1751.00	26.34
323 50 10 10 rep 3.txt	50	10	10	Solution	30.03	2624	2144.00	18.29
324 50 10 10 rep 4.txt	50	10	10	Solution	30.04	2322	1797.00	22.61
325 50 10 10 rep 5.txt	50	10	10	Solution	30.02	2467	1853.00	24.89
326 50 10 10 rep 6.txt	50	10	10	Solution	30.03	2695	1905.00	29.31
327 50 10 10 rep 7.txt	50	10	10	Solution	30.04	2542	2088.00	17.86
328 50 10 10 rep 8.txt	50	10	10	Solution	30.02	2442	2059.00	15.68
329 50 10 10 rep 9.txt	50	10	10	Solution	30.02	2283	1745.00	23.57
330 50 10 10 rep 10.txt	50	10	10	Solution	30.05	2686	1978.00	26.36
331 50 10 20 rep 1.txt	50	10	20	Solution	30.02	1288	1084.00	15.84
332 50 10 20 rep 2.txt	50	10	20	Solution	30.04	1261	1107.00	12.21
333 50 10 20 rep 3.txt	50	10	20	Solution	30.04	1399	1186.00	15.23
334 50 10 20 rep 4.txt	50	10	20	Solution	30.03	1301	1104.00	15.14
335 50 10 20 rep 5.txt	50	10	20	Solution	30.03	1232	1109.00	9.98
336 50 10 20 rep 6.txt	50	10	20	Solution	30.05	1251	1148.00	8.23
337 50 10 20 rep 7.txt	50	10	20	Solution	30.05	1287	1120.00	12.98
338 50 10 20 rep 8.txt	50	10	20	Solution	30.04	1250	1054.00	15.68
339 50 10 20 rep 9.txt	50	10	20	Solution	30.02	1453	1237.00	14.87
340 50 10 20 rep 10.txt	50	10	20	Solution	30.04	1135	950.00	16.30
341 50 10 30 rep 1.txt	50	10	30	Solution	30.04	946	851.00	10.04
342 50 10 30 rep 2.txt	50	10	30	Solution	30.03	944	817.00	13.45
343 50 10 30 rep 3.txt	50	10	30	Solution	30.03	982	869.00	11.51
344 50 10 30 rep 4.txt	50	10	30	Solution	30.04	953	826.00	13.33
345 50 10 30 rep 5.txt	50	10	30	Solution	30.02	992	867.00	12.60
346 50 10 30 rep 6.txt	50	10	30	Solution	30.02	1018	873.00	14.24
347 50 10 30 rep 7.txt	50	10	30	Solution	30.05	934	820.00	12.21
348 50 10 30 rep 8.txt	50	10	30	Solution	30.04	967	841.00	13.03
349 50 10 30 rep 9.txt	50	10	30	Solution	30.03	902	758.00	15.96
350 50 10 30 rep 10.txt	50	10	30	Solution	30.01	1001	888.00	11.29
351 50 10 50 rep 1.txt	50	10	50	Solution	30.05	741	708.00	4.45
352 50 10 50 rep 2.txt	50	10	50	Solution	30.05	683	663.00	2.93
353 50 10 50 rep 3.txt	50	10	50	Solution	30.05	744	705.00	5.24
354 50 10 50 rep 4.txt	50	10	50	Optimal	18.61	748	748.00	0.00
355 50 10 50 rep 5.txt	50	10	50	Optimal	1.82	791	791.00	0.00
356 50 10 50 rep 6.txt	50	10	50	Solution	30.06	735	691.00	5.99
357 50 10 50 rep 7.txt	50	10	50	Solution	30.03	729	690.00	5.35

Table 1: Results for CFS (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
358 50 10 50 rep 8.txt	50	10	50	Solution	30.04	714	713.00	0.14
359 50 10 50 rep 9.txt	50	10	50	Solution	30.02	750	696.00	7.20
360 50 10 50 rep 10.txt	50	10	50	Solution	30.04	747	691.00	7.50
361 100 3 10 rep 1.txt	100	3	10	Solution	30.04	3249	2997.00	7.76
362 100 3 10 rep 2.txt	100	3	10	Solution	30.03	3672	3166.00	13.78
363 100 3 10 rep 3.txt	100	3	10	Solution	30.03	3291	2973.00	9.66
364 100 3 10 rep 4.txt	100	3	10	Solution	30.03	3453	3035.00	12.11
365 100 3 10 rep 5.txt	100	3	10	Solution	30.03	3407	3166.00	7.07
366 100 3 10 rep 6.txt	100	3	10	Solution	30.03	3682	3237.00	12.09
367 100 3 10 rep 7.txt	100	3	10	Solution	30.04	3625	3212.00	11.39
368 100 3 10 rep 8.txt	100	3	10	Solution	30.02	3597	3275.00	8.95
369 100 3 10 rep 9.txt	100	3	10	Solution	30.02	3405	3013.00	11.51
370 100 3 10 rep 10.txt	100	3	10	Solution	30.02	3408	3065.00	10.06
371 100 3 20 rep 1.txt	100	3	20	Solution	30.02	1659	1516.00	8.62
372 100 3 20 rep 2.txt	100	3	20	Solution	30.04	1494	1382.00	7.50
373 100 3 20 rep 3.txt	100	3	20	Solution	30.02	1676	1489.00	11.16
374 100 3 20 rep 4.txt	100	3	20	Solution	30.02	1561	1405.00	9.99
375 100 3 20 rep 5.txt	100	3	20	Solution	30.01	1719	1657.00	3.61
376 100 3 20 rep 6.txt	100	3	20	Solution	30.02	1754	1599.00	8.84
377 100 3 20 rep 7.txt	100	3	20	Solution	30.05	1730	1576.00	8.90
378 100 3 20 rep 8.txt	100	3	20	Solution	30.03	1582	1476.00	6.70
379 100 3 20 rep 9.txt	100	3	20	Solution	30.02	1670	1580.00	5.39
380 100 3 20 rep 10.txt	100	3	20	Solution	30.03	1511	1401.00	7.28
381 100 3 30 rep 1.txt	100	3	30	Solution	30.02	1085	1001.00	7.74
382 100 3 30 rep 2.txt	100	3	30	Solution	30.02	1130	1076.00	4.78
383 100 3 30 rep 3.txt	100	3	30	Solution	30.03	1041	1018.00	2.21
384 100 3 30 rep 4.txt	100	3	30	Solution	30.05	1155	1100.00	4.76
385 100 3 30 rep 5.txt	100	3	30	Solution	30.04	1278	1225.00	4.15
386 100 3 30 rep 6.txt	100	3	30	Solution	30.03	1011	964.00	4.65
387 100 3 30 rep 7.txt	100	3	30	Solution	30.04	1122	1060.00	5.53
388 100 3 30 rep 8.txt	100	3	30	Solution	30.01	1083	1038.00	4.16
389 100 3 30 rep 9.txt	100	3	30	Solution	30.03	1095	1069.00	2.37
390 100 3 30 rep 10.txt	100	3	30	Solution	30.04	1106	1083.00	2.08
391 100 3 50 rep 1.txt	100	3	50	Solution	30.03	565	550.00	2.65
392 100 3 50 rep 2.txt	100	3	50	Solution	30.02	632	609.00	3.64
393 100 3 50 rep 3.txt	100	3	50	Solution	30.05	641	612.00	4.52
394 100 3 50 rep 4.txt	100	3	50	Solution	30.04	732	713.00	2.60
395 100 3 50 rep 5.txt	100	3	50	Solution	30.01	644	607.00	5.75
396 100 3 50 rep 6.txt	100	3	50	Solution	30.04	666	643.00	3.45
397 100 3 50 rep 7.txt	100	3	50	Solution	30.01	663	635.00	4.22
398 100 3 50 rep 8.txt	100	3	50	Solution	30.01	660	640.00	3.03
399 100 3 50 rep 9.txt	100	3	50	Solution	30.02	623	605.00	2.89
400 100 3 50 rep 10.txt	100	3	50	Solution	30.01	655	613.00	6.41
401 100 5 10 rep 1.txt	100	5	10	Solution	30.06	3671	2911.00	20.70
402 100 5 10 rep 2.txt	100	5	10	Solution	30.05	3787	3284.00	13.28

Table 1: Results for CFS (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
403 100 5 10 rep 3.txt	100	5	10	Solution	30.05	4079	3371.00	17.36
404 100 5 10 rep 4.txt	100	5	10	Solution	30.08	3717	3109.00	16.36
405 100 5 10 rep 5.txt	100	5	10	Solution	30.05	4075	3203.00	21.40
406 100 5 10 rep 6.txt	100	5	10	Solution	30.04	3770	3250.00	13.79
407 100 5 10 rep 7.txt	100	5	10	Solution	30.06	3794	3105.00	18.16
408 100 5 10 rep 8.txt	100	5	10	Solution	30.05	4046	3228.00	20.22
409 100 5 10 rep 9.txt	100	5	10	Solution	30.04	4024	2945.00	26.81
410 100 5 10 rep 10.txt	100	5	10	Solution	30.05	3888	3383.00	12.99
411 100 5 20 rep 1.txt	100	5	20	Solution	30.06	1989	1820.00	8.50
412 100 5 20 rep 2.txt	100	5	20	Solution	30.05	1761	1555.00	11.70
413 100 5 20 rep 3.txt	100	5	20	Solution	30.03	1931	1645.00	14.81
414 100 5 20 rep 4.txt	100	5	20	Solution	30.03	1788	1613.00	9.79
415 100 5 20 rep 5.txt	100	5	20	Solution	30.04	1948	1642.00	15.71
416 100 5 20 rep 6.txt	100	5	20	Solution	30.03	1721	1543.00	10.34
417 100 5 20 rep 7.txt	100	5	20	Solution	30.03	1980	1796.00	9.29
418 100 5 20 rep 8.txt	100	5	20	Solution	30.02	1717	1505.00	12.35
419 100 5 20 rep 9.txt	100	5	20	Solution	30.04	1663	1408.00	15.33
420 100 5 20 rep 10.txt	100	5	20	Solution	30.06	1912	1702.00	10.98
421 100 5 30 rep 1.txt	100	5	30	Solution	30.04	1229	1095.00	10.90
422 100 5 30 rep 2.txt	100	5	30	Solution	30.03	1218	1081.00	11.25
423 100 5 30 rep 3.txt	100	5	30	Solution	30.02	1241	1166.00	6.04
424 100 5 30 rep 4.txt	100	5	30	Solution	30.03	1175	1065.00	9.36
425 100 5 30 rep 5.txt	100	5	30	Solution	30.04	1183	1082.00	8.54
426 100 5 30 rep 6.txt	100	5	30	Solution	30.04	1201	1120.00	6.74
427 100 5 30 rep 7.txt	100	5	30	Solution	30.02	1267	1104.00	12.87
428 100 5 30 rep 8.txt	100	5	30	Solution	30.05	1272	1144.00	10.06
429 100 5 30 rep 9.txt	100	5	30	Solution	30.02	1241	1151.00	7.25
430 100 5 30 rep 10.txt	100	5	30	Solution	30.05	1180	1044.00	11.53
431 100 5 50 rep 1.txt	100	5	50	Solution	30.06	734	703.00	4.22
432 100 5 50 rep 2.txt	100	5	50	Solution	30.04	701	644.00	8.13
433 100 5 50 rep 3.txt	100	5	50	Solution	30.09	708	661.00	6.64
434 100 5 50 rep 4.txt	100	5	50	Solution	30.07	740	685.00	7.43
435 100 5 50 rep 5.txt	100	5	50	Solution	30.03	779	746.00	4.24
436 100 5 50 rep 6.txt	100	5	50	Solution	30.05	699	644.00	7.87
437 100 5 50 rep 7.txt	100	5	50	Solution	30.03	763	682.00	10.62
438 100 5 50 rep 8.txt	100	5	50	Solution	30.06	688	596.00	13.37
439 100 5 50 rep 9.txt	100	5	50	Solution	30.04	727	675.00	7.15
440 100 5 50 rep 10.txt	100	5	50	Solution	30.06	772	688.00	10.88
441 100 10 10 rep 1.txt	100	10	10	Solution	30.11	4530	3377.00	25.45
442 100 10 10 rep 2.txt	100	10	10	Solution	30.10	4697	3150.00	32.94
443 100 10 10 rep 3.txt	100	10	10	Solution	30.08	4665	3281.00	29.67
444 100 10 10 rep 4.txt	100	10	10	Solution	30.15	4632	3654.00	21.11
445 100 10 10 rep 5.txt	100	10	10	Solution	30.14	5438	3447.00	36.61
446 100 10 10 rep 6.txt	100	10	10	Solution	30.09	4618	3217.00	30.34
447 100 10 10 rep 7.txt	100	10	10	Solution	30.11	4728	3485.00	26.29

Table 1: Results for CFS (CPO) (480 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
448 100 10 10 rep 8.txt	100	10	10	Solution	30.12	5053	3527.00	30.20
449 100 10 10 rep 9.txt	100	10	10	Solution	30.14	4771	3435.00	28.00
450 100 10 10 rep 10.txt	100	10	10	Solution	30.14	4880	3851.00	21.09
451 100 10 20 rep 1.txt	100	10	20	Solution	30.05	2171	1897.00	12.62
452 100 10 20 rep 2.txt	100	10	20	Solution	30.07	2188	1825.00	16.59
453 100 10 20 rep 3.txt	100	10	20	Solution	30.11	2553	2209.00	13.47
454 100 10 20 rep 4.txt	100	10	20	Solution	30.07	2242	1797.00	19.85
455 100 10 20 rep 5.txt	100	10	20	Solution	30.06	2347	1971.00	16.02
456 100 10 20 rep 6.txt	100	10	20	Solution	30.12	2192	1656.00	24.45
457 100 10 20 rep 7.txt	100	10	20	Solution	30.09	2254	1746.00	22.54
458 100 10 20 rep 8.txt	100	10	20	Solution	30.05	2297	1958.00	14.76
459 100 10 20 rep 9.txt	100	10	20	Solution	30.06	2130	1834.00	13.90
460 100 10 20 rep 10.txt	100	10	20	Solution	30.13	2212	1786.00	19.26
461 100 10 30 rep 1.txt	100	10	30	Solution	30.05	1596	1283.00	19.61
462 100 10 30 rep 2.txt	100	10	30	Solution	30.07	1537	1258.00	18.15
463 100 10 30 rep 3.txt	100	10	30	Solution	30.15	1662	1425.00	14.26
464 100 10 30 rep 4.txt	100	10	30	Solution	30.18	1572	1263.00	19.66
465 100 10 30 rep 5.txt	100	10	30	Solution	30.07	1542	1294.00	16.08
466 100 10 30 rep 6.txt	100	10	30	Solution	30.19	1663	1283.00	22.85
467 100 10 30 rep 7.txt	100	10	30	Solution	30.07	1593	1357.00	14.81
468 100 10 30 rep 8.txt	100	10	30	Solution	30.06	1585	1313.00	17.16
469 100 10 30 rep 9.txt	100	10	30	Solution	30.04	1634	1342.00	17.87
470 100 10 30 rep 10.txt	100	10	30	Solution	30.18	1603	1297.00	19.09
471 100 10 50 rep 1.txt	100	10	50	Solution	30.14	1048	923.00	11.93
472 100 10 50 rep 2.txt	100	10	50	Solution	30.08	1041	900.00	13.54
473 100 10 50 rep 3.txt	100	10	50	Solution	30.12	1101	914.00	16.98
474 100 10 50 rep 4.txt	100	10	50	Solution	30.21	1021	910.00	10.87
475 100 10 50 rep 5.txt	100	10	50	Solution	30.05	1093	893.00	18.30
476 100 10 50 rep 6.txt	100	10	50	Solution	30.10	1076	947.00	11.99
477 100 10 50 rep 7.txt	100	10	50	Solution	30.04	1066	912.00	14.45
478 100 10 50 rep 8.txt	100	10	50	Solution	30.09	1046	895.00	14.44
479 100 10 50 rep 9.txt	100	10	50	Solution	30.12	1067	856.00	19.78
480 100 10 50 rep 10.txt	100	10	50	Solution	30.08	1031	898.00	12.90

2 Instances (CPSat)

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
1 20 3 10 rep 1.txt	20	3	10	Optimal	300.02	745	745.00	0.00
2 20 3 10 rep 2.txt	20	3	10	Optimal	300.15	739	739.00	0.00

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
3 20 3 10 rep 3.txt	20	3	10	Solution	300.10	494	480.00	2.83
4 20 3 10 rep 4.txt	20	3	10	Optimal	16.52	742	742.00	0.00
5 20 3 10 rep 5.txt	20	3	10	Solution	300.11	882	872.00	1.13
6 20 3 10 rep 6.txt	20	3	10	Optimal	226.90	726	726.00	0.00
7 20 3 10 rep 7.txt	20	3	10	Solution	300.11	715	705.00	1.40
8 20 3 10 rep 8.txt	20	3	10	Solution	300.09	874	824.00	5.72
9 20 3 10 rep 9.txt	20	3	10	Solution	300.09	593	567.00	4.38
10 20 3 10 rep 10.txt	20	3	10	Solution	300.55	784	781.00	0.38
11 20 3 20 rep 1.txt	20	3	20	Optimal	4.38	412	412.00	0.00
12 20 3 20 rep 2.txt	20	3	20	Optimal	300.06	358	358.00	0.00
13 20 3 20 rep 3.txt	20	3	20	Solution	300.25	420	414.00	1.43
14 20 3 20 rep 4.txt	20	3	20	Optimal	300.08	385	385.00	0.00
15 20 3 20 rep 5.txt	20	3	20	Optimal	300.08	350	350.00	0.00
16 20 3 20 rep 6.txt	20	3	20	Optimal	2.93	383	383.00	0.00
17 20 3 20 rep 7.txt	20	3	20	Optimal	300.06	372	372.00	0.00
18 20 3 20 rep 8.txt	20	3	20	Optimal	37.37	413	413.00	0.00
19 20 3 20 rep 9.txt	20	3	20	Optimal	300.04	297	297.00	0.00
20 20 3 20 rep 10.txt	20	3	20	Optimal	5.88	353	353.00	0.00
21 20 3 30 rep 1.txt	20	3	30	Optimal	300.04	335	335.00	0.00
22 20 3 30 rep 2.txt	20	3	30	Optimal	300.02	296	296.00	0.00
23 20 3 30 rep 3.txt	20	3	30	Optimal	39.23	304	304.00	0.00
24 20 3 30 rep 4.txt	20	3	30	Optimal	49.64	292	292.00	0.00
25 20 3 30 rep 5.txt	20	3	30	Optimal	300.02	292	292.00	0.00
26 20 3 30 rep 6.txt	20	3	30	Optimal	300.05	293	293.00	0.00
27 20 3 30 rep 7.txt	20	3	30	Optimal	300.05	345	345.00	0.00
28 20 3 30 rep 8.txt	20	3	30	Optimal	8.53	324	324.00	0.00
29 20 3 30 rep 9.txt	20	3	30	Optimal	300.03	300	300.00	0.00
30 20 3 30 rep 10.txt	20	3	30	Optimal	1.33	245	245.00	0.00
31 20 3 50 rep 1.txt	20	3	50	Optimal	0.03	253	253.00	0.00
32 20 3 50 rep 2.txt	20	3	50	Optimal	300.04	237	237.00	0.00
33 20 3 50 rep 3.txt	20	3	50	Optimal	0.02	241	241.00	0.00
34 20 3 50 rep 4.txt	20	3	50	Optimal	2.57	232	232.00	0.00
35 20 3 50 rep 5.txt	20	3	50	Optimal	0.02	272	272.00	0.00
36 20 3 50 rep 6.txt	20	3	50	Optimal	0.02	216	216.00	0.00
37 20 3 50 rep 7.txt	20	3	50	Optimal	0.01	224	224.00	0.00
38 20 3 50 rep 8.txt	20	3	50	Optimal	0.01	264	264.00	0.00
39 20 3 50 rep 9.txt	20	3	50	Optimal	3.44	234	234.00	0.00
40 20 3 50 rep 10.txt	20	3	50	Optimal	0.15	231	231.00	0.00
41 20 5 10 rep 1.txt	20	5	10	Solution	300.10	932	869.00	6.76
42 20 5 10 rep 2.txt	20	5	10	Solution	300.11	884	875.00	1.02
43 20 5 10 rep 3.txt	20	5	10	Solution	300.07	916	858.00	6.33
44 20 5 10 rep 4.txt	20	5	10	Solution	300.09	886	819.00	7.56
45 20 5 10 rep 5.txt	20	5	10	Solution	300.10	875	843.00	3.66
46 20 5 10 rep 6.txt	20	5	10	Optimal	300.02	832	832.00	0.00
47 20 5 10 rep 7.txt	20	5	10	Solution	300.08	947	826.00	12.78

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
48 20 5 10 rep 8.txt	20	5	10	Solution	300.18	918	839.00	8.61
49 20 5 10 rep 9.txt	20	5	10	Optimal	300.03	894	894.00	0.00
50 20 5 10 rep 10.txt	20	5	10	Solution	300.11	1084	1068.00	1.48
51 20 5 20 rep 1.txt	20	5	20	Optimal	300.03	497	497.00	0.00
52 20 5 20 rep 2.txt	20	5	20	Optimal	300.04	441	441.00	0.00
53 20 5 20 rep 3.txt	20	5	20	Solution	300.10	517	496.00	4.06
54 20 5 20 rep 4.txt	20	5	20	Solution	300.64	387	374.00	3.36
55 20 5 20 rep 5.txt	20	5	20	Optimal	300.08	478	478.00	0.00
56 20 5 20 rep 6.txt	20	5	20	Optimal	300.05	505	505.00	0.00
57 20 5 20 rep 7.txt	20	5	20	Optimal	300.05	466	466.00	0.00
58 20 5 20 rep 8.txt	20	5	20	Optimal	300.05	511	511.00	0.00
59 20 5 20 rep 9.txt	20	5	20	Optimal	300.03	522	522.00	0.00
60 20 5 20 rep 10.txt	20	5	20	Solution	300.09	481	460.00	4.37
61 20 5 30 rep 1.txt	20	5	30	Optimal	300.05	387	387.00	0.00
62 20 5 30 rep 2.txt	20	5	30	Optimal	300.03	452	452.00	0.00
63 20 5 30 rep 3.txt	20	5	30	Optimal	300.05	367	367.00	0.00
64 20 5 30 rep 4.txt	20	5	30	Optimal	300.04	427	427.00	0.00
65 20 5 30 rep 5.txt	20	5	30	Optimal	13.35	385	385.00	0.00
66 20 5 30 rep 6.txt	20	5	30	Optimal	1.44	402	402.00	0.00
67 20 5 30 rep 7.txt	20	5	30	Optimal	0.03	431	431.00	0.00
68 20 5 30 rep 8.txt	20	5	30	Optimal	0.54	404	404.00	0.00
69 20 5 30 rep 9.txt	20	5	30	Optimal	300.05	441	441.00	0.00
70 20 5 30 rep 10.txt	20	5	30	Optimal	300.04	437	437.00	0.00
71 20 5 50 rep 1.txt	20	5	50	Optimal	0.12	312	312.00	0.00
72 20 5 50 rep 2.txt	20	5	50	Optimal	0.06	323	323.00	0.00
73 20 5 50 rep 3.txt	20	5	50	Optimal	0.03	353	353.00	0.00
74 20 5 50 rep 4.txt	20	5	50	Optimal	0.02	348	348.00	0.00
75 20 5 50 rep 5.txt	20	5	50	Optimal	0.02	371	371.00	0.00
76 20 5 50 rep 6.txt	20	5	50	Optimal	0.02	425	425.00	0.00
77 20 5 50 rep 7.txt	20	5	50	Optimal	0.02	349	349.00	0.00
78 20 5 50 rep 8.txt	20	5	50	Optimal	0.01	398	398.00	0.00
79 20 5 50 rep 9.txt	20	5	50	Optimal	0.02	398	398.00	0.00
80 20 5 50 rep 10.txt	20	5	50	Optimal	0.03	386	386.00	0.00
81 20 10 10 rep 1.txt	20	10	10	Solution	300.24	1240	1045.00	15.73
82 20 10 10 rep 2.txt	20	10	10	Solution	300.08	1204	1049.00	12.87
83 20 10 10 rep 3.txt	20	10	10	Solution	300.09	1275	1156.00	9.33
84 20 10 10 rep 4.txt	20	10	10	Solution	300.11	1260	1180.00	6.35
85 20 10 10 rep 5.txt	20	10	10	Solution	300.31	1288	1178.00	8.54
86 20 10 10 rep 6.txt	20	10	10	Solution	300.24	1263	1174.00	7.05
87 20 10 10 rep 7.txt	20	10	10	Solution	300.11	1206	1069.00	11.36
88 20 10 10 rep 8.txt	20	10	10	Solution	300.09	1248	1198.00	4.01
89 20 10 10 rep 9.txt	20	10	10	Solution	300.10	1187	1087.00	8.42
90 20 10 10 rep 10.txt	20	10	10	Solution	300.14	1234	1156.00	6.32
91 20 10 20 rep 1.txt	20	10	20	Optimal	300.04	743	743.00	0.00
92 20 10 20 rep 2.txt	20	10	20	Solution	300.07	831	776.00	6.62

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
93 20 10 20 rep 3.txt	20	10	20	Optimal	300.03	704	704.00	0.00
94 20 10 20 rep 4.txt	20	10	20	Solution	300.12	833	763.00	8.40
95 20 10 20 rep 5.txt	20	10	20	Solution	300.09	732	710.00	3.01
96 20 10 20 rep 6.txt	20	10	20	Solution	300.12	871	835.00	4.13
97 20 10 20 rep 7.txt	20	10	20	Optimal	300.04	798	798.00	0.00
98 20 10 20 rep 8.txt	20	10	20	Solution	300.13	799	765.00	4.26
99 20 10 20 rep 9.txt	20	10	20	Solution	300.10	792	729.00	7.95
100 20 10 20 rep 10.txt	20	10	20	Optimal	300.03	747	747.00	0.00
101 20 10 30 rep 1.txt	20	10	30	Optimal	300.04	730	730.00	0.00
102 20 10 30 rep 2.txt	20	10	30	Optimal	0.36	780	780.00	0.00
103 20 10 30 rep 3.txt	20	10	30	Optimal	300.04	615	615.00	0.00
104 20 10 30 rep 4.txt	20	10	30	Optimal	0.29	726	726.00	0.00
105 20 10 30 rep 5.txt	20	10	30	Optimal	300.04	667	667.00	0.00
106 20 10 30 rep 6.txt	20	10	30	Optimal	72.25	639	639.00	0.00
107 20 10 30 rep 7.txt	20	10	30	Optimal	0.51	728	728.00	0.00
108 20 10 30 rep 8.txt	20	10	30	Optimal	0.27	763	763.00	0.00
109 20 10 30 rep 9.txt	20	10	30	Optimal	24.41	645	645.00	0.00
110 20 10 30 rep 10.txt	20	10	30	Optimal	0.17	692	692.00	0.00
111 20 10 50 rep 1.txt	20	10	50	Optimal	0.05	668	668.00	0.00
112 20 10 50 rep 2.txt	20	10	50	Optimal	0.04	740	740.00	0.00
113 20 10 50 rep 3.txt	20	10	50	Optimal	0.04	701	701.00	0.00
114 20 10 50 rep 4.txt	20	10	50	Optimal	0.03	693	693.00	0.00
115 20 10 50 rep 5.txt	20	10	50	Optimal	0.04	651	651.00	0.00
116 20 10 50 rep 6.txt	20	10	50	Optimal	0.08	641	641.00	0.00
117 20 10 50 rep 7.txt	20	10	50	Optimal	0.03	654	654.00	0.00
118 20 10 50 rep 8.txt	20	10	50	Optimal	0.04	673	673.00	0.00
119 20 10 50 rep 9.txt	20	10	50	Optimal	0.03	708	708.00	0.00
120 20 10 50 rep 10.txt	20	10	50	Optimal	0.06	731	731.00	0.00
121 30 3 10 rep 1.txt	30	3	10	Solution	300.12	993	897.00	9.67
122 30 3 10 rep 2.txt	30	3	10	Solution	300.25	1249	1237.00	0.96
123 30 3 10 rep 3.txt	30	3	10	Solution	300.13	1010	907.00	10.20
124 30 3 10 rep 4.txt	30	3	10	Solution	300.31	1202	1199.00	0.25
125 30 3 10 rep 5.txt	30	3	10	Optimal	1.82	1105	1105.00	0.00
126 30 3 10 rep 6.txt	30	3	10	Solution	300.11	1203	1198.00	0.42
127 30 3 10 rep 7.txt	30	3	10	Optimal	300.03	1221	1221.00	0.00
128 30 3 10 rep 8.txt	30	3	10	Optimal	300.08	1181	1181.00	0.00
129 30 3 10 rep 9.txt	30	3	10	Solution	300.39	1058	972.00	8.13
130 30 3 10 rep 10.txt	30	3	10	Optimal	300.07	1039	1039.00	0.00
131 30 3 20 rep 1.txt	30	3	20	Solution	300.16	559	526.00	5.90
132 30 3 20 rep 2.txt	30	3	20	Solution	300.19	573	543.00	5.24
133 30 3 20 rep 3.txt	30	3	20	Solution	300.16	486	449.00	7.61
134 30 3 20 rep 4.txt	30	3	20	Solution	300.26	619	594.00	4.04
135 30 3 20 rep 5.txt	30	3	20	Solution	300.07	466	422.00	9.44
136 30 3 20 rep 6.txt	30	3	20	Solution	300.22	497	488.00	1.81
137 30 3 20 rep 7.txt	30	3	20	Solution	300.21	502	474.00	5.58

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
138 30 3 20 rep 8.txt	30	3	20	Solution	300.19	553	537.00	2.89
139 30 3 20 rep 9.txt	30	3	20	Solution	300.19	497	486.00	2.21
140 30 3 20 rep 10.txt	30	3	20	Solution	300.14	466	455.00	2.36
141 30 3 30 rep 1.txt	30	3	30	Solution	300.09	438	425.00	2.97
142 30 3 30 rep 2.txt	30	3	30	Solution	300.07	324	317.00	2.16
143 30 3 30 rep 3.txt	30	3	30	Solution	300.21	446	440.00	1.35
144 30 3 30 rep 4.txt	30	3	30	Solution	300.15	341	333.00	2.35
145 30 3 30 rep 5.txt	30	3	30	Solution	300.16	432	425.00	1.62
146 30 3 30 rep 6.txt	30	3	30	Solution	300.13	327	309.00	5.50
147 30 3 30 rep 7.txt	30	3	30	Solution	300.16	412	407.00	1.21
148 30 3 30 rep 8.txt	30	3	30	Solution	300.08	389	382.00	1.80
149 30 3 30 rep 9.txt	30	3	30	Solution	300.15	416	412.00	0.96
150 30 3 30 rep 10.txt	30	3	30	Solution	300.19	363	359.00	1.10
151 30 3 50 rep 1.txt	30	3	50	Optimal	0.51	252	252.00	0.00
152 30 3 50 rep 2.txt	30	3	50	Optimal	300.04	258	258.00	0.00
153 30 3 50 rep 3.txt	30	3	50	Optimal	7.40	294	294.00	0.00
154 30 3 50 rep 4.txt	30	3	50	Optimal	300.03	235	235.00	0.00
155 30 3 50 rep 5.txt	30	3	50	Optimal	300.04	253	253.00	0.00
156 30 3 50 rep 6.txt	30	3	50	Optimal	300.04	268	268.00	0.00
157 30 3 50 rep 7.txt	30	3	50	Optimal	300.03	253	253.00	0.00
158 30 3 50 rep 8.txt	30	3	50	Optimal	12.21	260	260.00	0.00
159 30 3 50 rep 9.txt	30	3	50	Optimal	0.40	275	275.00	0.00
160 30 3 50 rep 10.txt	30	3	50	Solution	300.13	264	259.00	1.89
161 30 5 10 rep 1.txt	30	5	10	Solution	300.09	1288	1206.00	6.37
162 30 5 10 rep 2.txt	30	5	10	Solution	300.13	1231	1179.00	4.22
163 30 5 10 rep 3.txt	30	5	10	Solution	300.10	1281	1168.00	8.82
164 30 5 10 rep 4.txt	30	5	10	Solution	300.09	1155	1122.00	2.86
165 30 5 10 rep 5.txt	30	5	10	Solution	300.30	1337	1330.00	0.52
166 30 5 10 rep 6.txt	30	5	10	Solution	300.44	1214	1098.00	9.56
167 30 5 10 rep 7.txt	30	5	10	Solution	300.08	1222	1182.00	3.27
168 30 5 10 rep 8.txt	30	5	10	Solution	300.16	1683	1541.00	8.44
169 30 5 10 rep 9.txt	30	5	10	Solution	300.09	1039	1022.00	1.64
170 30 5 10 rep 10.txt	30	5	10	Solution	300.08	1442	1195.00	17.13
171 30 5 20 rep 1.txt	30	5	20	Solution	300.15	675	584.00	13.48
172 30 5 20 rep 2.txt	30	5	20	Solution	300.15	618	587.00	5.02
173 30 5 20 rep 3.txt	30	5	20	Solution	300.17	692	621.00	10.26
174 30 5 20 rep 4.txt	30	5	20	Solution	300.49	680	644.00	5.29
175 30 5 20 rep 5.txt	30	5	20	Solution	300.18	582	535.00	8.08
176 30 5 20 rep 6.txt	30	5	20	Solution	300.54	673	656.00	2.53
177 30 5 20 rep 7.txt	30	5	20	Solution	300.12	659	606.00	8.04
178 30 5 20 rep 8.txt	30	5	20	Solution	300.13	641	556.00	13.26
179 30 5 20 rep 9.txt	30	5	20	Solution	300.09	620	544.00	12.26
180 30 5 20 rep 10.txt	30	5	20	Solution	300.08	686	655.00	4.52
181 30 5 30 rep 1.txt	30	5	30	Solution	300.10	454	432.00	4.85
182 30 5 30 rep 2.txt	30	5	30	Solution	300.08	471	448.00	4.88

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
183 30 5 30 rep 3.txt	30	5	30	Solution	300.14	487	456.00	6.37
184 30 5 30 rep 4.txt	30	5	30	Solution	300.08	485	454.00	6.39
185 30 5 30 rep 5.txt	30	5	30	Solution	300.09	509	469.00	7.86
186 30 5 30 rep 6.txt	30	5	30	Solution	300.13	412	399.00	3.16
187 30 5 30 rep 7.txt	30	5	30	Solution	300.13	468	449.00	4.06
188 30 5 30 rep 8.txt	30	5	30	Solution	300.08	469	460.00	1.92
189 30 5 30 rep 9.txt	30	5	30	Solution	300.14	473	455.00	3.81
190 30 5 30 rep 10.txt	30	5	30	Solution	300.08	511	470.00	8.02
191 30 5 50 rep 1.txt	30	5	50	Optimal	300.03	368	368.00	0.00
192 30 5 50 rep 2.txt	30	5	50	Optimal	0.30	378	378.00	0.00
193 30 5 50 rep 3.txt	30	5	50	Optimal	300.04	379	379.00	0.00
194 30 5 50 rep 4.txt	30	5	50	Optimal	148.98	382	382.00	0.00
195 30 5 50 rep 5.txt	30	5	50	Optimal	300.03	367	367.00	0.00
196 30 5 50 rep 6.txt	30	5	50	Optimal	300.02	379	379.00	0.00
197 30 5 50 rep 7.txt	30	5	50	Optimal	8.78	394	394.00	0.00
198 30 5 50 rep 8.txt	30	5	50	Optimal	0.05	445	445.00	0.00
199 30 5 50 rep 9.txt	30	5	50	Optimal	2.09	360	360.00	0.00
200 30 5 50 rep 10.txt	30	5	50	Optimal	300.04	378	378.00	0.00
201 30 10 10 rep 1.txt	30	10	10	Solution	300.10	1889	1605.00	15.03
202 30 10 10 rep 2.txt	30	10	10	Solution	300.10	1562	1195.00	23.50
203 30 10 10 rep 3.txt	30	10	10	Solution	300.08	1632	1304.00	20.10
204 30 10 10 rep 4.txt	30	10	10	Solution	300.08	1729	1497.00	13.42
205 30 10 10 rep 5.txt	30	10	10	Solution	300.15	1636	1309.00	19.99
206 30 10 10 rep 6.txt	30	10	10	Solution	300.09	1735	1544.00	11.01
207 30 10 10 rep 7.txt	30	10	10	Solution	300.08	1628	1368.00	15.97
208 30 10 10 rep 8.txt	30	10	10	Solution	300.10	1602	1300.00	18.85
209 30 10 10 rep 9.txt	30	10	10	Solution	300.15	1731	1329.00	23.22
210 30 10 10 rep 10.txt	30	10	10	Solution	300.15	1797	1457.00	18.92
211 30 10 20 rep 1.txt	30	10	20	Solution	300.10	961	826.00	14.05
212 30 10 20 rep 2.txt	30	10	20	Solution	300.11	881	729.00	17.25
213 30 10 20 rep 3.txt	30	10	20	Solution	300.11	970	865.00	10.82
214 30 10 20 rep 4.txt	30	10	20	Solution	300.15	907	795.00	12.35
215 30 10 20 rep 5.txt	30	10	20	Solution	300.13	923	852.00	7.69
216 30 10 20 rep 6.txt	30	10	20	Solution	300.14	986	895.00	9.23
217 30 10 20 rep 7.txt	30	10	20	Solution	300.10	1013	869.00	14.22
218 30 10 20 rep 8.txt	30	10	20	Solution	300.09	934	867.00	7.17
219 30 10 20 rep 9.txt	30	10	20	Solution	300.13	880	791.00	10.11
220 30 10 20 rep 10.txt	30	10	20	Solution	300.14	940	833.00	11.38
221 30 10 30 rep 1.txt	30	10	30	Solution	300.08	753	725.00	3.72
222 30 10 30 rep 2.txt	30	10	30	Solution	300.11	782	742.00	5.12
223 30 10 30 rep 3.txt	30	10	30	Optimal	300.04	698	698.00	0.00
224 30 10 30 rep 4.txt	30	10	30	Solution	300.09	723	710.00	1.80
225 30 10 30 rep 5.txt	30	10	30	Solution	300.08	763	702.00	7.99
226 30 10 30 rep 6.txt	30	10	30	Solution	300.13	794	731.00	7.93
227 30 10 30 rep 7.txt	30	10	30	Solution	300.10	734	696.00	5.18

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
228 30 10 30 rep 8.txt	30	10	30	Optimal	300.04	791	791.00	0.00
229 30 10 30 rep 9.txt	30	10	30	Solution	300.08	751	711.00	5.33
230 30 10 30 rep 10.txt	30	10	30	Solution	300.09	771	754.00	2.20
231 30 10 50 rep 1.txt	30	10	50	Optimal	0.11	738	738.00	0.00
232 30 10 50 rep 2.txt	30	10	50	Optimal	0.18	646	646.00	0.00
233 30 10 50 rep 3.txt	30	10	50	Optimal	0.15	753	753.00	0.00
234 30 10 50 rep 4.txt	30	10	50	Optimal	0.97	652	652.00	0.00
235 30 10 50 rep 5.txt	30	10	50	Optimal	0.22	667	667.00	0.00
236 30 10 50 rep 6.txt	30	10	50	Optimal	0.12	687	687.00	0.00
237 30 10 50 rep 7.txt	30	10	50	Optimal	0.21	827	827.00	0.00
238 30 10 50 rep 8.txt	30	10	50	Optimal	0.19	692	692.00	0.00
239 30 10 50 rep 9.txt	30	10	50	Optimal	0.37	671	671.00	0.00
240 30 10 50 rep 10.txt	30	10	50	Optimal	0.14	764	764.00	0.00
241 50 3 10 rep 1.txt	50	3	10	Solution	300.08	1860	1722.00	7.42
242 50 3 10 rep 2.txt	50	3	10	Solution	300.11	1676	1637.00	2.33
243 50 3 10 rep 3.txt	50	3	10	Solution	300.13	1841	1686.00	8.42
244 50 3 10 rep 4.txt	50	3	10	Solution	300.09	1822	1753.00	3.79
245 50 3 10 rep 5.txt	50	3	10	Solution	300.16	1663	1582.00	4.87
246 50 3 10 rep 6.txt	50	3	10	Solution	300.09	1790	1618.00	9.61
247 50 3 10 rep 7.txt	50	3	10	Solution	300.13	1675	1592.00	4.96
248 50 3 10 rep 8.txt	50	3	10	Solution	300.13	1644	1489.00	9.43
249 50 3 10 rep 9.txt	50	3	10	Solution	300.13	1820	1713.00	5.88
250 50 3 10 rep 10.txt	50	3	10	Solution	300.10	1485	1363.00	8.22
251 50 3 20 rep 1.txt	50	3	20	Solution	300.14	769	721.00	6.24
252 50 3 20 rep 2.txt	50	3	20	Solution	300.15	898	851.00	5.23
253 50 3 20 rep 3.txt	50	3	20	Solution	300.15	813	764.00	6.03
254 50 3 20 rep 4.txt	50	3	20	Solution	300.15	770	712.00	7.53
255 50 3 20 rep 5.txt	50	3	20	Solution	300.10	806	740.00	8.19
256 50 3 20 rep 6.txt	50	3	20	Solution	300.18	855	801.00	6.32
257 50 3 20 rep 7.txt	50	3	20	Solution	300.16	892	814.00	8.74
258 50 3 20 rep 8.txt	50	3	20	Solution	300.16	822	781.00	4.99
259 50 3 20 rep 9.txt	50	3	20	Solution	300.11	807	761.00	5.70
260 50 3 20 rep 10.txt	50	3	20	Solution	300.12	786	735.00	6.49
261 50 3 30 rep 1.txt	50	3	30	Solution	300.10	567	539.00	4.94
262 50 3 30 rep 2.txt	50	3	30	Solution	300.10	666	637.00	4.35
263 50 3 30 rep 3.txt	50	3	30	Solution	300.11	529	505.00	4.54
264 50 3 30 rep 4.txt	50	3	30	Solution	300.09	538	510.00	5.20
265 50 3 30 rep 5.txt	50	3	30	Solution	300.11	580	563.00	2.93
266 50 3 30 rep 6.txt	50	3	30	Solution	300.12	572	529.00	7.52
267 50 3 30 rep 7.txt	50	3	30	Solution	300.16	617	605.00	1.94
268 50 3 30 rep 8.txt	50	3	30	Solution	301.10	568	538.00	5.28
269 50 3 30 rep 9.txt	50	3	30	Solution	300.09	619	591.00	4.52
270 50 3 30 rep 10.txt	50	3	30	Solution	300.12	595	577.00	3.03
271 50 3 50 rep 1.txt	50	3	50	Solution	300.12	369	349.00	5.42
272 50 3 50 rep 2.txt	50	3	50	Solution	300.10	369	354.00	4.07

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
273 50 3 50 rep 3.txt	50	3	50	Solution	300.07	348	333.00	4.31
274 50 3 50 rep 4.txt	50	3	50	Solution	300.10	380	372.00	2.11
275 50 3 50 rep 5.txt	50	3	50	Solution	300.07	362	337.00	6.91
276 50 3 50 rep 6.txt	50	3	50	Solution	300.07	381	362.00	4.99
277 50 3 50 rep 7.txt	50	3	50	Solution	300.07	383	369.00	3.66
278 50 3 50 rep 8.txt	50	3	50	Solution	300.11	335	326.00	2.69
279 50 3 50 rep 9.txt	50	3	50	Solution	300.06	380	358.00	5.79
280 50 3 50 rep 10.txt	50	3	50	Solution	300.07	358	350.00	2.23
281 50 5 10 rep 1.txt	50	5	10	Solution	300.11	1845	1671.00	9.43
282 50 5 10 rep 2.txt	50	5	10	Solution	300.12	2066	1949.00	5.66
283 50 5 10 rep 3.txt	50	5	10	Solution	300.20	1801	1637.00	9.11
284 50 5 10 rep 4.txt	50	5	10	Solution	300.09	2054	1831.00	10.86
285 50 5 10 rep 5.txt	50	5	10	Solution	300.11	2205	1944.00	11.84
286 50 5 10 rep 6.txt	50	5	10	Solution	300.09	2023	1833.00	9.39
287 50 5 10 rep 7.txt	50	5	10	Solution	300.09	1916	1896.00	1.04
288 50 5 10 rep 8.txt	50	5	10	Solution	300.12	1912	1602.00	16.21
289 50 5 10 rep 9.txt	50	5	10	Solution	300.10	2063	1972.00	4.41
290 50 5 10 rep 10.txt	50	5	10	Solution	300.09	1919	1803.00	6.04
291 50 5 20 rep 1.txt	50	5	20	Solution	300.15	971	917.00	5.56
292 50 5 20 rep 2.txt	50	5	20	Solution	300.11	960	856.00	10.83
293 50 5 20 rep 3.txt	50	5	20	Solution	300.10	960	857.00	10.73
294 50 5 20 rep 4.txt	50	5	20	Solution	300.12	1114	975.00	12.48
295 50 5 20 rep 5.txt	50	5	20	Solution	300.14	1051	964.00	8.28
296 50 5 20 rep 6.txt	50	5	20	Solution	300.10	966	896.00	7.25
297 50 5 20 rep 7.txt	50	5	20	Solution	300.19	915	883.00	3.50
298 50 5 20 rep 8.txt	50	5	20	Solution	300.09	941	860.00	8.61
299 50 5 20 rep 9.txt	50	5	20	Solution	300.11	945	852.00	9.84
300 50 5 20 rep 10.txt	50	5	20	Solution	300.10	907	871.00	3.97
301 50 5 30 rep 1.txt	50	5	30	Solution	300.09	647	602.00	6.96
302 50 5 30 rep 2.txt	50	5	30	Solution	300.09	665	602.00	9.47
303 50 5 30 rep 3.txt	50	5	30	Solution	300.07	648	589.00	9.10
304 50 5 30 rep 4.txt	50	5	30	Solution	300.10	670	615.00	8.21
305 50 5 30 rep 5.txt	50	5	30	Solution	300.07	670	628.00	6.27
306 50 5 30 rep 6.txt	50	5	30	Solution	300.08	695	657.00	5.47
307 50 5 30 rep 7.txt	50	5	30	Solution	300.06	701	628.00	10.41
308 50 5 30 rep 8.txt	50	5	30	Solution	300.09	741	719.00	2.97
309 50 5 30 rep 9.txt	50	5	30	Solution	300.10	664	616.00	7.23
310 50 5 30 rep 10.txt	50	5	30	Solution	300.09	714	682.00	4.48
311 50 5 50 rep 1.txt	50	5	50	Solution	300.07	437	401.00	8.24
312 50 5 50 rep 2.txt	50	5	50	Solution	300.07	479	458.00	4.38
313 50 5 50 rep 3.txt	50	5	50	Solution	300.07	470	445.00	5.32
314 50 5 50 rep 4.txt	50	5	50	Solution	300.07	455	417.00	8.35
315 50 5 50 rep 5.txt	50	5	50	Solution	300.10	475	438.00	7.79
316 50 5 50 rep 6.txt	50	5	50	Solution	300.06	418	397.00	5.02
317 50 5 50 rep 7.txt	50	5	50	Optimal	300.03	440	440.00	0.00

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
318 50 5 50 rep 8.txt	50	5	50	Solution	300.08	441	406.00	7.94
319 50 5 50 rep 9.txt	50	5	50	Solution	300.06	429	403.00	6.06
320 50 5 50 rep 10.txt	50	5	50	Solution	300.08	426	390.00	8.45
321 50 10 10 rep 1.txt	50	10	10	Solution	300.09	2426	2070.00	14.67
322 50 10 10 rep 2.txt	50	10	10	Solution	300.08	2319	1818.00	21.60
323 50 10 10 rep 3.txt	50	10	10	Solution	300.10	2549	2296.00	9.93
324 50 10 10 rep 4.txt	50	10	10	Solution	300.09	2279	1810.00	20.58
325 50 10 10 rep 5.txt	50	10	10	Solution	300.08	2377	1985.00	16.49
326 50 10 10 rep 6.txt	50	10	10	Solution	300.09	2637	2057.00	21.99
327 50 10 10 rep 7.txt	50	10	10	Solution	300.10	2559	2241.00	12.43
328 50 10 10 rep 8.txt	50	10	10	Solution	300.14	2474	2121.00	14.27
329 50 10 10 rep 9.txt	50	10	10	Solution	300.20	2270	1860.00	18.06
330 50 10 10 rep 10.txt	50	10	10	Solution	300.08	2763	2025.00	26.71
331 50 10 20 rep 1.txt	50	10	20	Solution	300.08	1260	1092.00	13.33
332 50 10 20 rep 2.txt	50	10	20	Solution	300.13	1257	1109.00	11.77
333 50 10 20 rep 3.txt	50	10	20	Solution	300.11	1392	1189.00	14.58
334 50 10 20 rep 4.txt	50	10	20	Solution	300.18	1288	1120.00	13.04
335 50 10 20 rep 5.txt	50	10	20	Solution	300.11	1246	1114.00	10.59
336 50 10 20 rep 6.txt	50	10	20	Solution	300.10	1272	1159.00	8.88
337 50 10 20 rep 7.txt	50	10	20	Solution	300.09	1236	1125.00	8.98
338 50 10 20 rep 8.txt	50	10	20	Solution	300.09	1215	1055.00	13.17
339 50 10 20 rep 9.txt	50	10	20	Solution	300.12	1427	1241.00	13.03
340 50 10 20 rep 10.txt	50	10	20	Solution	300.10	1126	954.00	15.28
341 50 10 30 rep 1.txt	50	10	30	Solution	300.12	950	853.00	10.21
342 50 10 30 rep 2.txt	50	10	30	Solution	300.08	940	819.00	12.87
343 50 10 30 rep 3.txt	50	10	30	Solution	300.08	990	878.00	11.31
344 50 10 30 rep 4.txt	50	10	30	Solution	300.11	950	837.00	11.89
345 50 10 30 rep 5.txt	50	10	30	Solution	300.12	978	861.00	11.96
346 50 10 30 rep 6.txt	50	10	30	Solution	300.09	1028	877.00	14.69
347 50 10 30 rep 7.txt	50	10	30	Solution	300.10	936	820.00	12.39
348 50 10 30 rep 8.txt	50	10	30	Solution	300.08	970	842.00	13.20
349 50 10 30 rep 9.txt	50	10	30	Solution	300.09	876	762.00	13.01
350 50 10 30 rep 10.txt	50	10	30	Solution	300.12	952	889.00	6.62
351 50 10 50 rep 1.txt	50	10	50	Solution	300.11	734	708.00	3.54
352 50 10 50 rep 2.txt	50	10	50	Solution	300.08	682	663.00	2.79
353 50 10 50 rep 3.txt	50	10	50	Solution	300.07	745	705.00	5.37
354 50 10 50 rep 4.txt	50	10	50	Optimal	300.05	748	748.00	0.00
355 50 10 50 rep 5.txt	50	10	50	Optimal	300.04	791	791.00	0.00
356 50 10 50 rep 6.txt	50	10	50	Solution	300.08	725	692.00	4.55
357 50 10 50 rep 7.txt	50	10	50	Solution	300.08	738	690.00	6.50
358 50 10 50 rep 8.txt	50	10	50	Optimal	300.03	713	713.00	0.00
359 50 10 50 rep 9.txt	50	10	50	Solution	300.08	750	694.00	7.47
360 50 10 50 rep 10.txt	50	10	50	Solution	300.12	745	691.00	7.25
361 100 3 10 rep 1.txt	100	3	10	Solution	300.12	3221	3198.00	0.71
362 100 3 10 rep 2.txt	100	3	10	Solution	300.18	3748	3332.00	11.10

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
363 100 3 10 rep 3.txt	100	3	10	Solution	300.09	3315	3014.00	9.08
364 100 3 10 rep 4.txt	100	3	10	Solution	300.10	3450	3201.00	7.22
365 100 3 10 rep 5.txt	100	3	10	Solution	300.09	3411	3290.00	3.55
366 100 3 10 rep 6.txt	100	3	10	Solution	300.21	3655	3463.00	5.25
367 100 3 10 rep 7.txt	100	3	10	Solution	300.10	3621	3451.00	4.69
368 100 3 10 rep 8.txt	100	3	10	Solution	300.08	3588	3438.00	4.18
369 100 3 10 rep 9.txt	100	3	10	Solution	300.20	3480	2964.00	14.83
370 100 3 10 rep 10.txt	100	3	10	Solution	300.10	3378	3234.00	4.26
371 100 3 20 rep 1.txt	100	3	20	Solution	300.14	1691	1518.00	10.23
372 100 3 20 rep 2.txt	100	3	20	Solution	300.11	1515	1381.00	8.84
373 100 3 20 rep 3.txt	100	3	20	Solution	300.17	1661	1490.00	10.30
374 100 3 20 rep 4.txt	100	3	20	Solution	300.11	1558	1409.00	9.56
375 100 3 20 rep 5.txt	100	3	20	Solution	300.15	1693	1657.00	2.13
376 100 3 20 rep 6.txt	100	3	20	Solution	300.13	1752	1601.00	8.62
377 100 3 20 rep 7.txt	100	3	20	Solution	300.12	1696	1581.00	6.78
378 100 3 20 rep 8.txt	100	3	20	Solution	300.09	1588	1481.00	6.74
379 100 3 20 rep 9.txt	100	3	20	Solution	300.13	1667	1582.00	5.10
380 100 3 20 rep 10.txt	100	3	20	Solution	300.10	1529	1401.00	8.37
381 100 3 30 rep 1.txt	100	3	30	Solution	300.11	1060	1001.00	5.57
382 100 3 30 rep 2.txt	100	3	30	Solution	300.13	1128	1075.00	4.70
383 100 3 30 rep 3.txt	100	3	30	Solution	300.14	1042	1018.00	2.30
384 100 3 30 rep 4.txt	100	3	30	Solution	300.08	1170	1101.00	5.90
385 100 3 30 rep 5.txt	100	3	30	Solution	300.09	1286	1228.00	4.51
386 100 3 30 rep 6.txt	100	3	30	Solution	300.08	1023	964.00	5.77
387 100 3 30 rep 7.txt	100	3	30	Solution	300.10	1115	1061.00	4.84
388 100 3 30 rep 8.txt	100	3	30	Solution	300.11	1101	1046.00	5.00
389 100 3 30 rep 9.txt	100	3	30	Solution	300.10	1102	1069.00	2.99
390 100 3 30 rep 10.txt	100	3	30	Solution	300.08	1112	1086.00	2.34
391 100 3 50 rep 1.txt	100	3	50	Solution	300.07	578	550.00	4.84
392 100 3 50 rep 2.txt	100	3	50	Solution	300.08	628	611.00	2.71
393 100 3 50 rep 3.txt	100	3	50	Solution	300.07	657	622.00	5.33
394 100 3 50 rep 4.txt	100	3	50	Solution	300.09	732	713.00	2.60
395 100 3 50 rep 5.txt	100	3	50	Solution	300.09	656	609.00	7.16
396 100 3 50 rep 6.txt	100	3	50	Solution	300.08	680	651.00	4.26
397 100 3 50 rep 7.txt	100	3	50	Solution	300.12	666	636.00	4.50
398 100 3 50 rep 8.txt	100	3	50	Solution	300.09	666	643.00	3.45
399 100 3 50 rep 9.txt	100	3	50	Solution	300.12	624	606.00	2.88
400 100 3 50 rep 10.txt	100	3	50	Solution	300.08	654	619.00	5.35
401 100 5 10 rep 1.txt	100	5	10	Solution	300.12	3721	2746.00	26.20
402 100 5 10 rep 2.txt	100	5	10	Solution	300.19	3758	2679.00	28.71
403 100 5 10 rep 3.txt	100	5	10	Solution	300.21	4146	3447.00	16.86
404 100 5 10 rep 4.txt	100	5	10	Solution	300.12	3684	3159.00	14.25
405 100 5 10 rep 5.txt	100	5	10	Solution	300.13	3930	2687.00	31.63
406 100 5 10 rep 6.txt	100	5	10	Solution	300.17	3805	3028.00	20.42
407 100 5 10 rep 7.txt	100	5	10	Solution	300.21	3708	2369.00	36.11

Table 2: Results for CFS (CPSat) (420 Instances)

Name	Nr Jobs	Nr Machines	Capa	Status	Time	Makespan	Bound	Gap Percent
408 100 5 10 rep 8.txt	100	5	10	Solution	300.17	4020	2478.00	38.36
409 100 5 10 rep 9.txt	100	5	10	Solution	300.15	4050	2751.00	32.07
410 100 5 10 rep 10.txt	100	5	10	Solution	300.19	3921	3302.00	15.79
411 100 5 20 rep 1.txt	100	5	20	Solution	300.21	2024	1822.00	9.98
412 100 5 20 rep 2.txt	100	5	20	Solution	300.14	1744	1558.00	10.67
413 100 5 20 rep 3.txt	100	5	20	Solution	300.10	1902	1645.00	13.51
414 100 5 20 rep 4.txt	100	5	20	Solution	300.11	1766	1615.00	8.55
415 100 5 20 rep 5.txt	100	5	20	Solution	300.09	1943	1649.00	15.13
416 100 5 20 rep 6.txt	100	5	20	Solution	300.10	1740	1543.00	11.32
417 100 5 20 rep 7.txt	100	5	20	Solution	300.15	1972	1801.00	8.67
418 100 5 20 rep 8.txt	100	5	20	Solution	300.10	1686	1505.00	10.74
419 100 5 20 rep 9.txt	100	5	20	Solution	300.13	1640	1412.00	13.90
420 100 5 20 rep 10.txt	100	5	20	Solution	300.13	1945	1706.00	12.29

3 Comparison

Table 3: Comparison of CPO and CPSat for Results of CFS

Group	Nr	All Instances Optimal (% of All Instances)				Optimal Only Time (% of VB)		Non Optimal Only Cost (% of VB)		Bound (% of VB)	
		Both	CPO	CPSat	None	CPO	CPSat	CPO	CPSat	CPO	CPSat
20/0/3/10	10	0.00	0.00	40.00	60.00	n/a	n/a	100.44	100.03	92.42	100.00
20/0/3/20	10	80.00	0.00	10.00	10.00	101.48	2398.81	100.13	100.00	98.73	100.00
20/0/3/30	10	100.00	0.00	0.00	0.00	100.00	5852.98	n/a	n/a	n/a	n/a
20/0/3/50	10	100.00	0.00	0.00	0.00	101.54	57492.08	n/a	n/a	n/a	n/a
20/0/5/10	10	0.00	0.00	20.00	80.00	n/a	n/a	102.78	100.00	89.82	100.00
20/0/5/20	10	40.00	0.00	30.00	30.00	100.00	2789.85	100.77	100.00	96.98	99.43
20/0/5/30	10	90.00	0.00	10.00	0.00	101.17	3875.71	100.70	100.00	98.59	100.00
20/0/5/50	10	100.00	0.00	0.00	0.00	102.63	168.04	n/a	n/a	n/a	n/a
20/0/10/10	10	0.00	0.00	0.00	100.00	n/a	n/a	103.12	100.00	91.06	100.00
20/0/10/20	10	20.00	0.00	20.00	60.00	100.00	2061.37	101.39	100.00	96.88	99.82
20/0/10/30	10	90.00	0.00	10.00	0.00	100.00	24958.00	100.15	100.00	99.25	100.00
20/0/10/50	10	100.00	0.00	0.00	0.00	100.00	223.43	n/a	n/a	n/a	n/a
30/0/3/10	10	0.00	0.00	40.00	60.00	n/a	n/a	100.33	100.07	91.39	100.00
30/0/3/20	10	0.00	0.00	0.00	100.00	n/a	n/a	102.18	100.00	99.44	100.00
30/0/3/30	10	0.00	0.00	0.00	100.00	n/a	n/a	101.11	100.00	99.66	99.97
30/0/3/50	10	90.00	0.00	0.00	10.00	100.00	4489.43	101.14	100.00	99.61	100.00
30/0/5/10	10	0.00	0.00	0.00	100.00	n/a	n/a	101.85	100.26	90.10	100.00
30/0/5/20	10	0.00	0.00	0.00	100.00	n/a	n/a	101.56	100.00	98.76	100.00
30/0/5/30	10	0.00	0.00	0.00	100.00	n/a	n/a	101.08	100.04	99.13	99.93
30/0/5/50	10	90.00	0.00	10.00	0.00	100.00	4350.68	100.00	100.00	99.46	100.00
30/0/10/10	10	0.00	0.00	0.00	100.00	n/a	n/a	101.52	100.48	94.62	100.00
30/0/10/20	10	0.00	0.00	0.00	100.00	n/a	n/a	101.85	100.31	99.12	99.96
30/0/10/30	10	10.00	0.00	10.00	80.00	100.00	47929.76	100.95	100.06	99.72	99.91
30/0/10/50	10	100.00	0.00	0.00	0.00	100.00	404.45	n/a	n/a	n/a	n/a
50/0/3/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.71	100.30	95.33	100.00
50/0/3/20	10	0.00	0.00	0.00	100.00	n/a	n/a	101.45	100.11	99.74	100.00
50/0/3/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.55	100.43	99.64	100.00
50/0/3/50	10	0.00	0.00	0.00	100.00	n/a	n/a	100.11	100.69	99.83	99.89
50/0/5/10	10	0.00	0.00	0.00	100.00	n/a	n/a	101.55	100.12	94.29	100.00
50/0/5/20	10	0.00	0.00	0.00	100.00	n/a	n/a	101.14	100.12	99.50	99.98
50/0/5/30	10	0.00	0.00	0.00	100.00	n/a	n/a	102.00	100.07	99.40	99.98
50/0/5/50	10	10.00	0.00	0.00	90.00	100.00	35050.77	100.45	100.17	100.00	99.58
50/0/10/10	10	0.00	0.00	0.00	100.00	n/a	n/a	101.46	100.51	94.81	100.00
50/0/10/20	10	0.00	0.00	0.00	100.00	n/a	n/a	101.36	100.28	99.47	100.00
50/0/10/30	10	0.00	0.00	0.00	100.00	n/a	n/a	101.01	100.28	99.60	99.93
50/0/10/50	10	20.00	0.00	10.00	70.00	100.00	2938.17	100.36	100.17	99.98	99.96
100/0/3/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.29	100.52	95.42	99.85
100/0/3/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.51	100.47	99.86	99.99
100/0/3/30	10	0.00	0.00	0.00	100.00	n/a	n/a	100.31	100.61	99.85	99.99
100/0/3/50	10	0.00	0.00	0.00	100.00	n/a	n/a	100.08	101.00	99.47	100.00
100/0/5/10	10	0.00	0.00	0.00	100.00	n/a	n/a	100.83	100.55	99.61	89.76
100/0/5/20	10	0.00	0.00	0.00	100.00	n/a	n/a	100.74	100.48	99.83	100.00
100/0/5/30	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/5/50	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/10/10	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/10/20	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/10/30	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
100/0/10/50	0	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a